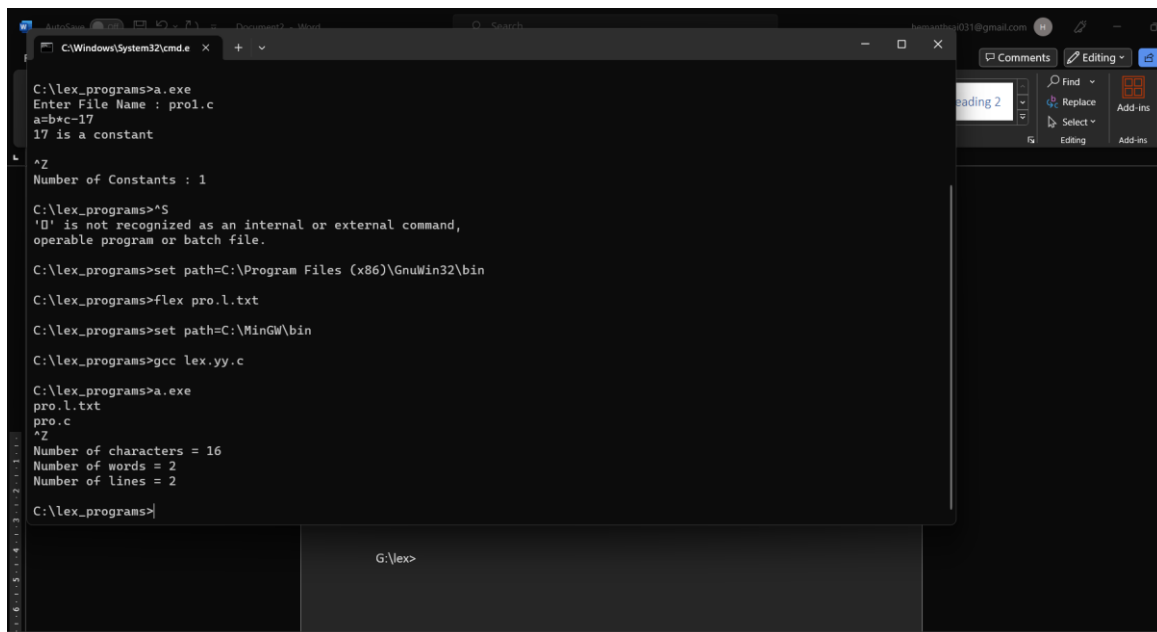


Write a LEX specification file to take input C program from a .c file and count the number of characters, number of lines & number of words.

**Input Source Program: (sample.c)**

```
#include <stdio.h>

int main()
{
    int number1, number2, sum;
    printf("Enter two integers: ");
    scanf("%d %d", &number1, &number2);
    sum = number1 + number2;
    printf("%d + %d = %d", number1, number2, sum);
    return 0;
}
```



```
C:\Windows\System32\cmd.exe
C:\lex_programs>a.exe
Enter File Name : pro1.c
a=b*c-17
17 is a constant
^Z
Number of Constants : 1

C:\lex_programs>^S
'D' is not recognized as an internal or external command,
operable program or batch file.

C:\lex_programs>set path=C:\Program Files (x86)\GnuWin32\bin

C:\lex_programs>flex pro.l.txt

C:\lex_programs>set path=C:\MinGW\bin

C:\lex_programs>gcc lex.yy.c

C:\lex_programs>a.exe
pro.l.txt
pro.c
^Z
Number of characters = 16
Number of words = 2
Number of lines = 2

C:\lex_programs>
```

**Program: (count\_lines.l)**

```
%{
int nchar, nword, nline;
}%
%%
\n { nline++; nchar++; }
[^ \t \n]+ { nword++; nchar += yytext; }
. { nchar++; }
%%
int yywrap(void) {
return 1;
}
```

```
}  
int main(int argc, char *argv[]) {  
    yyin = fopen(argv[1], "r");  
    yylex();  
    printf("Number of characters = %d\n", nchar);  
    printf("Number of words = %d\n", nword);  
    printf("Number of lines = %d\n", nline);  
    fclose(yyin);  
}
```

### **Output:**

G:\lex>flex count\_line.l

G:\lex>gcc lex.yy.c

G:\lex>a.exe sample.c

Number of characters = 233

Number of words = 33

Number of lines = 10

G:\lex>